Roseway Labs KTP: Class-Incremental Learning Evaluation

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Model Performance & Retention

Experiment Overview

This dashboard tracks the Class-Incremental Learning (CIL) performance of a DistilBERT model trained across 100 food categories (10 incremental stages).

To mitigate catastrophic forgetting while adhering to strict memory constraints, this pipeline utilises **Parameter-Efficient Fine-Tuning (LoRA)** combined with an **Experience Replay** memory buffer.

Final Performance Metrics

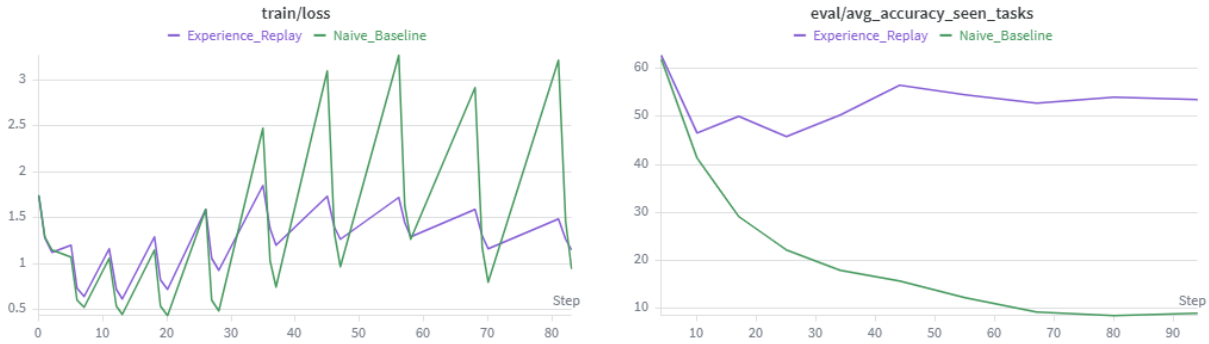
The primary quantitative measure of forgetting used is **Backward Transfer (BWT)**:

$$BWT = \frac{1}{T-1} \sum_{i=1}^{T-1} (R_{T,i} - R_{i,i}) \tag{1}$$

Table 1: Final Continual Learning Metrics

Metric	Naïve Baseline	Experience Replay
Final Avg. Accuracy	7.92%	54.15%
Backward Transfer	-61.29%	-17.73%

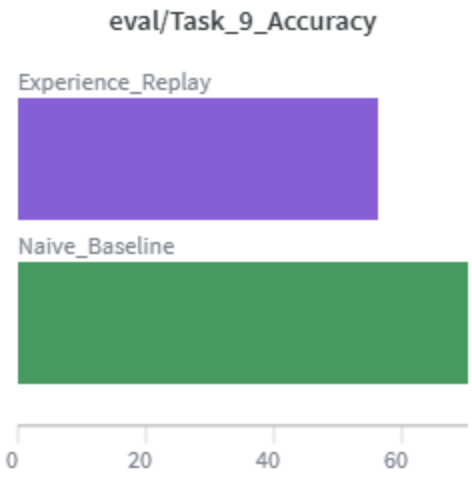
The Naïve Baseline suffered total catastrophic forgetting, while Experience Replay demonstrated robust knowledge retention.



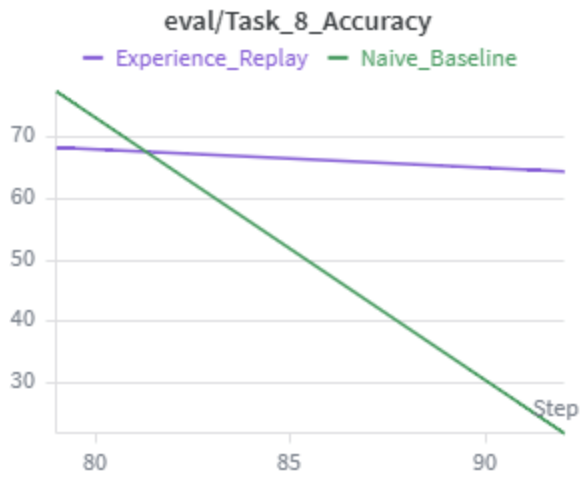
(a) Training Loss

(b) Average Accuracy on Seen Tasks

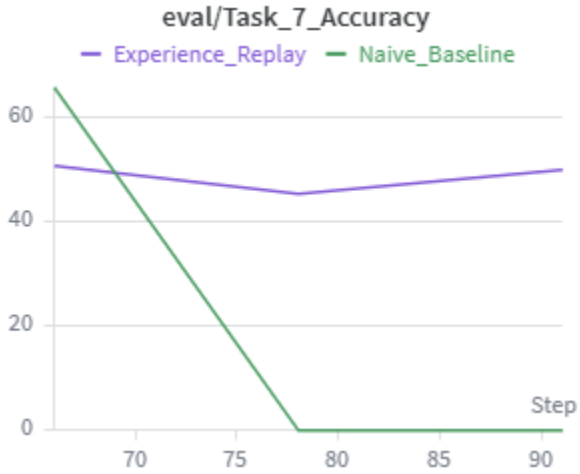
Figure 1: Overall Model Performance across 10 incremental stages.



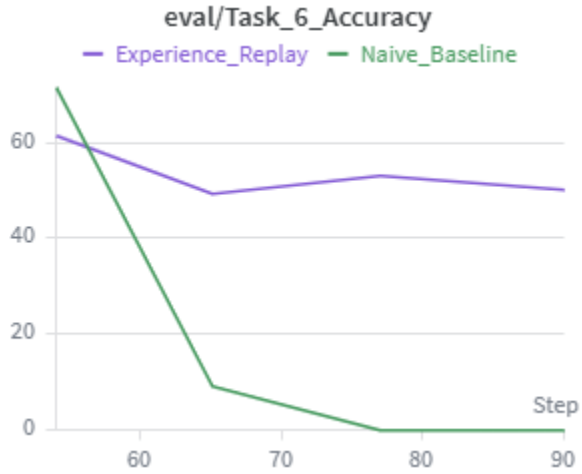
(a) Task 9



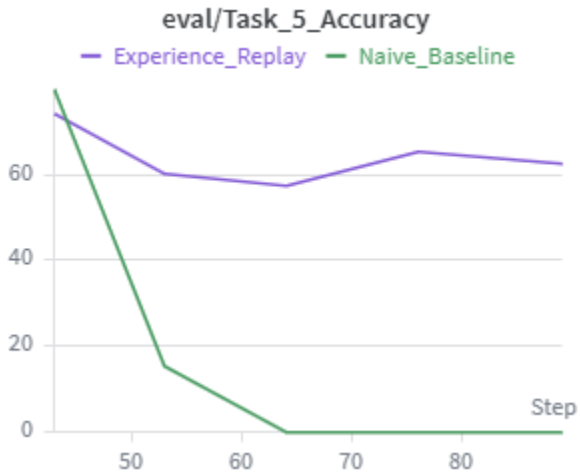
(b) Task 8



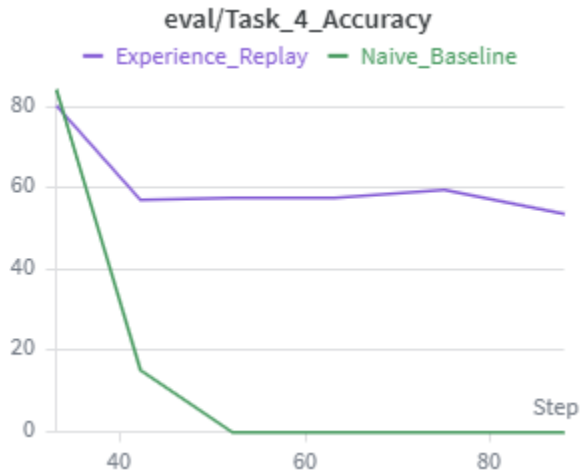
(c) Task 7



(d) Task 6

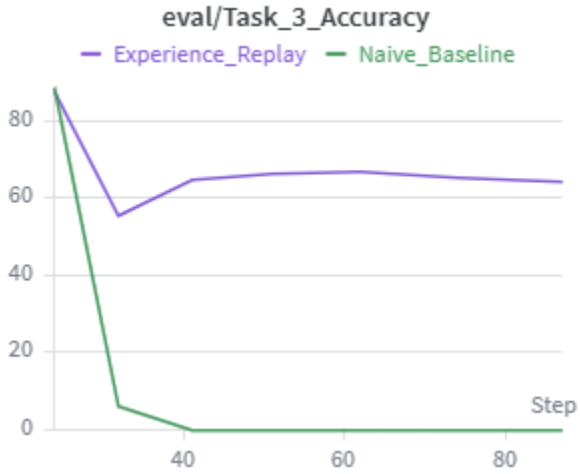


(e) Task 5

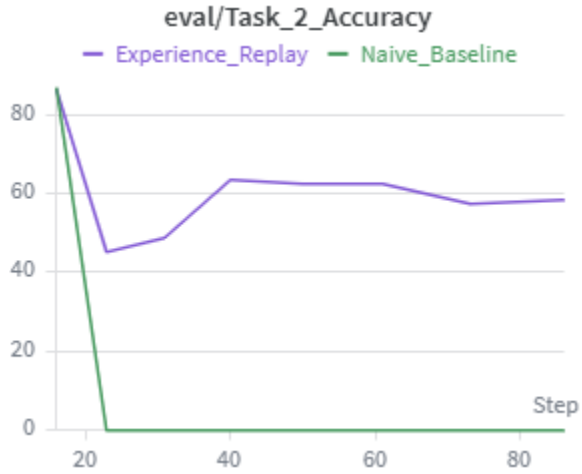


(f) Task 4

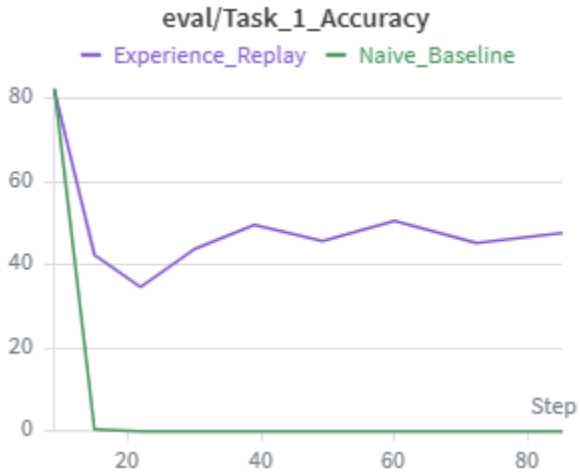
Figure 2: Individual Accuracy Progression per Task (Part 1).



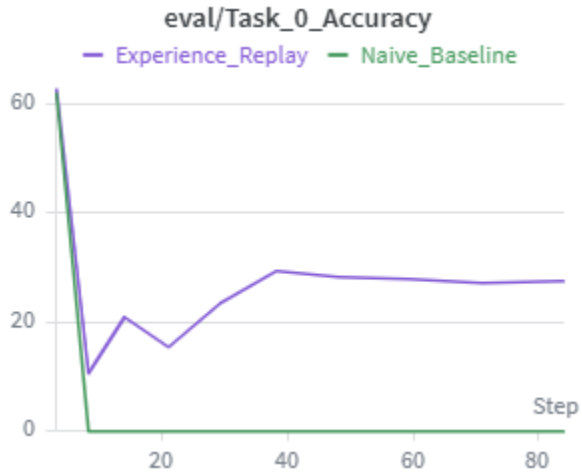
(g) Task 3



(h) Task 2



(i) Task 1



(j) Task 0

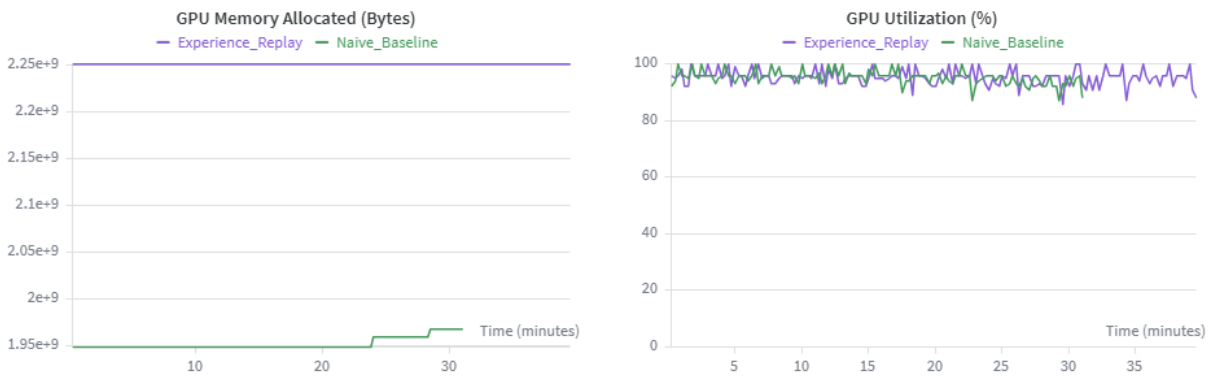
Figure 2: Individual Accuracy Progression per Task (Part 2). Note the organic recovery of historical tasks (e.g., Task 0 and Task 1) due to Experience Replay.

Hardware & Computational Efficiency

In a live compounding pharmacy setting, the ability to rapidly iterate daily outweighs marginal accuracy gains that require days of compute time.

By freezing the pre-trained backbone and injecting trainable low-rank matrices (LoRA), trainable parameters were restricted to ∼1.1 million.

- **Training Speed:** $<$ 30 minutes per stage on a standard T4 GPU.
- **Memory Footprint:** Bounded GPU VRAM usage to ∼1.6 GB, ensuring deployability on edge-tier hardware.



(a) Allocated Memory (Bytes) (b) Utilization (%)

Figure 3: GPU Performance Metrics during the training cycle.